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MARKSCHEME

SPECIMEN

CHEMISTRY

Standard Level

Paper 3

17 pages

Subject Details: Chemistry SL Paper 3 Markscheme

Mark Allocation

Candidates are required to answer **ALL** questions in Section A **[15 marks]** and all questions from **ONE** option in Section B **[20 marks]**.
Maximum total = **[35 marks]**.

1. Each row in the "Question" column relates to the smallest subpart of the question.

2. The maximum mark for each question subpart is indicated in the "Total" column.

3. Each marking point in the "Answers" column is shown by means of a tick (✓) at the end of the marking point.

4. A question subpart may have more marking points than the total allows. This will be indicated by "**max**" written after the mark in the "Total" column. The related rubric, if necessary, will be outlined in the "Notes" column.

5. An alternative wording is indicated in the "Answers" column by a slash (/). Either wording can be accepted.

6. An alternative answer is indicated in the "Answers" column by '***OR***' on the line between the alternatives. Either answer can be accepted.

7. Words in angled brackets ‹ › in the "Answers" column are not necessary to gain the mark.

8. Words that are <u>underlined</u> are essential for the mark.

9. The order of marking points does not have to be as in the "Answers" column, unless stated otherwise in the "Notes" column.

10. If the candidate's answer has the same "meaning" or can be clearly interpreted as being of equivalent significance, detail and validity as that in the "Answers" column then award the mark. Where this point is considered to be particularly relevant in a question it is emphasized by ***OWTTE*** (or words to that effect) in the "Notes" column.

11. Remember that many candidates are writing in a second language. Effective communication is more important than grammatical accuracy.

12. Occasionally, a part of a question may require an answer that is required for subsequent marking points. If an error is made in the first marking point then it should be penalized. However, if the incorrect answer is used correctly in subsequent marking points then **follow through** marks should be awarded. When marking, indicate this by adding **ECF** (error carried forward) on the script. "ECF acceptable" will be displayed in the "Notes" column.

13. Do **not** penalize candidates for errors in units or significant figures, **unless** it is specifically referred to in the "Notes" column.

14. If a question specifically asks for the name of a substance, do not award a mark for a correct formula unless directed otherwise in the "Notes" column, similarly, if the formula is specifically asked for, unless directed otherwise in the "Notes" column do not award a mark for a correct name.

15. If a question asks for an equation for a reaction, a balanced symbol equation is usually expected, do not award a mark for a word equation or an unbalanced equation unless directed otherwise in the "Notes" column.

16. Ignore missing or incorrect state symbols in an equation unless directed otherwise in the "Notes" column.

SECTION A

Question			Answers	Notes	Total
1.	**a**	**i**	relative values ***OR*** compared with a standard ***OR*** not absolute measure ✓		**1**
	a	**ii**	high ODP for compounds with high Cl ***OR*** low ODP for compounds with less Cl ***OR*** zero ODP for compounds with no Cl ✓		**1**
	b		increasing atmospheric lifetime correlates with increasing GWP ✓ total contribution to global warming depends on length of time in atmosphere ***OR*** GWP depends on efficiency as greenhouse gas and atmospheric lifetime ✓	*Accept alternate answers based on sound scientific reasoning.*	**2**
	c	**i**	1,1,1,2-tetrafluoroethane ✓	*Allow without commas or dashes.*	**1**
	c	**ii**	$M\,(CH_2FCF_3) = (12.01\times 2) + (1.01\times 2) + (19.00\times 4) = 102.04\langle g\,mol^{-1}\rangle$ ✓ $\Delta H\,(\text{condensation } CH_2FCF_3) = -[0.217\langle kJ\,g^{-1}\rangle \times 102.04\langle g\,mol^{-1}\rangle] = -22.1\langle kJ\,mol^{-1}\rangle$ ✓	*Award **[1 max]** for* $\Delta H = 22.1\,kJ$	**2**
	c	**iii**	atmospheric lifetime CO_2 much longer than CH_2FCF_3 ***OR*** after 100 years approx 30 % CO_2 still present whereas CH_2FCF_3 removed ✓ CO_2 from current emissions will continue to effect climate change/global warming far into the future ✓	*OWTTE*	**2**

Question			Answers	Notes	Total
2.	**a**		forgot to take account of water of crystallisation ***OR*** should have used 24.972 g ✓	*OWTTE*	**1**
	b		less uncertainty in the volume ***OR*** more precise ✓ takes into account volume change on dissolving ***OR*** concentration is for a given volume of solution not volume of solvent ✓		**2**
	c		filter ***OR*** centrifuge ✓ rinse (the solid) with water ✓ heat in an oven ***OR*** rinse with propanone/ethanol/volatile organic solvent and leave to evaporate ✓	*Award **[2]** for all 3, **[1]** for any 2.*	**2**
	d		taking a known mass of the solid to react directly with glucose ***OR*** not making the standard solution ✓	*OWTTE* *Accept any other valid answer based on sound scientific reasoning.*	**1**

SECTION B

Option A — Materials

Question			Answers	Notes	Total
3.	**a**		rigid ***OR*** rod-shaped/long thin molecule ✓		**1**
	b	**i**	mixture of carbon containing compound and inert dilutant in gas/vapour phase ✓ passed over a heated metal catalyst ✓		**2**
	b	**ii**	(very) large surface area ✓		**1**

4.	**a**			*Award **[1]** per correct row or column.*	**2**

Compound	Magnesium oxide	Cobalt(II) oxide	
Electronegativity difference	2.1	1.5	✓
Average electronegativity	2.35	2.65	✓

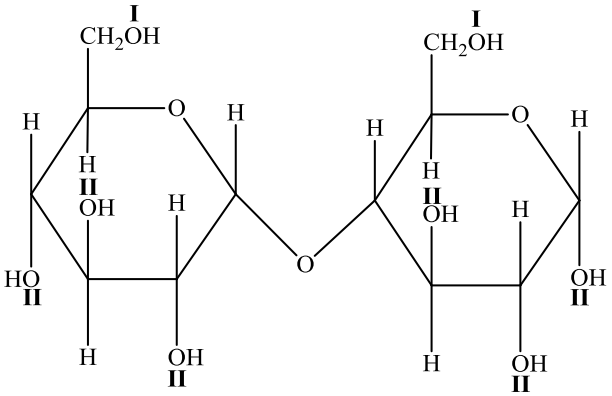
	b			*Award **[1]** per correct row or column.*	**2**

Compound	Magnesium oxide	Cobalt(II) oxide	
Bond type	Ionic	Polar covalent	✓
% covalent character	30 – 35	53 – 58	✓

Question			Answers	Notes	Total
5.	**a**		positive argon ions and (free) electrons ✓		**1**
	b	**i**	mol Mg^{2+} = ‹$0.25\times10\times10^{-6}$ =› 2.5×10^{-6} ‹mol› ✓ mass Mg^{2+} = ‹$24.31\times2.5\times10^{-6}$ =› 6.08×10^{-5} g ✓		**2**
	b	**ii**	mass of solid too small to weigh accurately ✓ successive dilution of solution ***OR*** dilution of concentrated solution ✓	*OWTTE*	**2**
	b	**iii**	627 kcps **and** it lies inside of the calibrated region ***OR*** 627 kcps **and** 12 kcps lies outside of calibrated region ✓	*Accept other correct suggestions, for example "low values such as 12 kcps would have very high uncertainty".*	**1**

Question			Answers	Notes	Total
6.	**a**		resin identification codes ✓ ensures uniformity for recycling ✓ ***OR*** addition/condensation ✓ classification into similar reaction types ✓ ***OR*** flexible ✓ direct towards appropriate uses ✓ ***OR*** brittle ✓ direct towards appropriate uses ✓	*OWTTE* *Accept "predict possible monomers". OWTTE* *Accept any other valid scientific classification with a justifiable scientific reason for **[2]**.*	**2**
	b		softens the polymer ✓ separates the polymer chains ***OR*** reduces intermolecular forces ✓		**2**
	c		all of the reagents end up in useful product ***OR*** atom economy is 100% ***OR*** there is no chemical waste ✓		**1**
	d		hydrogen chloride/HCl ***OR*** dioxin ✓		**1**

Option B — Biochemistry

Question			Answers	Notes	Total
7.	**a**		 ✓	*Award mark for a correctly placed **I** **and** a correctly placed **II**.* *Allow **II** placed on hemiacetal.*	**1**
	b	**i**	$C_{12}H_{22}O_{11} + H_2O \rightarrow 2C_6H_{12}O_6$ ✓		**1**
	b	**ii**	catabolism ✓	*Accept hydrolysis.*	**1**
	c		at X (low pH) enzyme/protein protonated/positively charged/cationic (so unable to bind effectively) ✓ at Y (optimum pH) enzyme maximally able to bind to substrate/maltose ✓ at Z (high pH) enzyme/protein deprotonated/negatively charged/anionic (so unable to bind effectively) ✓	*Award **[1 max]** for reference to denaturation/change in shape of active site without explanation in terms of changes in ionization.*	**3**
	d		R_f value $V = \frac{5.4}{5.9} = 0.91$ **and** R_f value $W = \frac{1.5}{5.9} = 0.25$ ✓ so W is glutamine (V cannot be identified) ✓		**2**

Question			Answers	Notes	Total
8.	**a**		$C_{18}H_{34}O_3$ ✓		**1**
	b	**i**	both have 18 carbon atoms ✓ both have COOH/carboxylic acid group ***OR*** both are fatty acids ✓ ricinoleic acid has a carbon-carbon double bond/C=C/‹mono›unsaturated whereas stearic acid has all single C–C bonds/saturated ✓ ricinoleic acid has an OH/hydroxyl group ‹in the chain› whereas stearic acid does not ✓	*Do not accept just acids in M2* *Any 3 for **[3 max]**.*	**3 max**
	b	**ii**	ricinoleic acid more likely to undergo oxidative rancidity ‹than stearic acid› ✓ carbon-carbon double bond/C=C can be oxidised ✓	*OWTTE*	**2**
	c	**i**	‹heating causes› denaturation ***OR*** ‹heating causes› loss of conformation ***OR*** ‹heating causes› change of shape ***OR*** ‹heating causes› inability to bind substrates ✓	*Do not accept inactivated.*	**1**
	c	**ii**	castor seeds contain toxins/ricin ***OR*** ingesting raw seeds can be fatal ✓ different health/safety standards in different countries ***OR*** richer countries exploit workers in less-developed/poorer countries ✓	*Accept alternate valid answers, such as economic considerations.*	**2**

Question			Answers	Notes	Total
9.	**a**		substance/chemical/compound found in organism not normally present ***OR*** compound foreign to living organism ✓	*Accept artificially synthesised/man-made compound in the environment/biosphere.*	**1**
	b		non-polar ***OR*** lipophilic ***OR*** structure based on phenyl/hydrocarbon ***OR*** hydrophobic interactions ***OR*** similar (non)polarity to fat ✓		**1**
	c		ionic bonds ✓ hydrogen bonds ✓ van der Waals' forces ✓ hydrophobic interactions ✓	*Award **[1]** for any 3 correct answers.* *Accept alternate valid answers other than covalent bonding.*	**1 max**

Option C — Energy

Question			Answers	Notes	Total
10.	**a**	**i**	viscosity too high ✓		**1**
	a	**ii**	alcohol **and** (strong) acid ***OR*** base ✓	*Accept any specific alcohol (eg ethanol).*	**1**
	b	**i**	fermentation ✓		**1**
	b	**ii**	‹$\frac{703\times5470}{114.26}$› ⇒ 33700 ‹$\mathrm{kJ\,dm^{-3}}$› ✓		**1**
	b	**iii**	more energy from a given volume of fuel ✓	*Accept greater energy density.*	**1**
	b	**iv**	ethanol: $\frac{1367}{2} = 683.5\,\mathrm{kJ\,mol^{-1}}$ **and** octane: $\frac{5470}{8} = 683.8\,\mathrm{kJ\,mol^{-1}}$ ***OR*** mass of CO_2 produced in the release of 1000 kJ ethanol: $\frac{2\times44.01\times1000}{1367} = 64.4\,\mathrm{g}$ **and** octane: $\frac{8\times44.01\times1000}{1367} = 64.4\,\mathrm{g}$ ✓	*Accept other methods that show the amount carbon dioxide produced for the same heat energy output is the same for both fuels.*	**1**
	b	**v**	ethanol is a biofuel/produced from plant material ***OR*** growing plants absorbs carbon dioxide ✓		**1**

Question			Answers	Notes	Total
11.	**a**		nuclear power benefitted from the race to develop nuclear weapons ✓	*OWTTE* *Accept other valid explanations.*	**1**
	b		$\Delta m = \langle 234.99333 - 135.90722 - 89.907738 - [9 \times 1.00867] = \rangle 0.100342 \langle \text{amu} \rangle$ ✓ $= \langle 0.100342 \times 1.66 \times 10^{-27} \langle \text{kg} \rangle = \rangle 1.67 \times 10^{-28} \langle \text{kg} \rangle$ ✓ $E = \langle mc^2 = 1.67 \times 10^{-28} \times (3 \times 10^8)^2 = \rangle 1.50 \times 10^{-11} \langle \text{J} \rangle$ ✓	*Award **[3]** for correct final answer.*	**3**
	c		*Short term:* in cooling ponds ✓ *Long term:* vitrification ***OR*** underground in stable geological formations ✓		**2**

Question			Answers	Notes	Total
12.	**a**		incoming solar radiation is short wavelength ***OR*** incoming solar radiation is high frequency ***OR*** incoming solar radiation is high energy radiation ***OR*** incoming solar radiation is (UV radiation) ✓ radiation emitted (by the Earth's surface) is long wavelength ***OR*** radiation emitted (by the Earth's surface) is low frequency ***OR*** radiation emitted (by the Earth's surface) is low energy ***OR*** radiation emitted (by the Earth's surface) is IR radiation ✓ this energy is absorbed in the bonds of greenhouse gases ***OR*** the molecules vibrate when IR radiation is absorbed ✓ this energy is then re-radiated ‹some of it towards the surface of the Earth› ✓		**3 max**
	b		stretching ***OR*** bending ✓ causing a change in polarity/dipole moment ✓		**2**
	c	**i**	methane **and** anaerobic decomposition of organic matter ***OR*** digestion in animals ✓	*Accept other examples of greenhouse gases with correct natural sources.*	**1**
	c	**ii**	major sources of water vapour are natural rather than anthropogenic/due to humans ***OR*** levels of water vapour have remained almost constant whereas those of CO_2 have increased significantly in recent times ✓		**1**

Option D — Medicinal chemistry

Question			Answers	Notes	Total
13.	**a**		esterification ***OR*** condensation ✓		**1**
	b		*Difference*: only spectrum for salicylic acid has ‹strong broad› peak from 3200–3600 cm^{-1} for OH ‹in alcohol/phenol› ✓ *Similarities*: both have ‹strong› peaks from 1050–1410 cm^{-1} for C–O ‹in alcohol/phenol› ✓ both have ‹strong› peaks from1700–1750 cm^{-1} for C=O ‹in carboxylic acid› ✓ both have ‹broad› peaks from 2500–3000 cm^{-1} for OH ‹in carboxylic acid› ✓ both have peaks from 2850–3090 cm^{-1} for C–H ✓	*Accept "acetylsalicylic acid has two peaks in the 1700–1800 cm^{-1} range due to 2 different C=O".* *Award **[2 max]** for two of the following similarities.*	**3 max**
	c		reaction with NaOH produces ‹ionic› salt ***OR*** $C_6H_4(OH)(COOH) + NaOH \rightarrow C_6H_4(OH)(COONa) + H_2O$ ✓ increases ‹aqueous› solubility ‹for transport/uptake› ✓ higher proportion of drug/dosage reaches target region/cells ✓		**3**

Question			Answers	Notes	Total
14.	**a**		lack cell structure ***OR*** exist within host cell ***OR*** mutate easily and frequently ✓		**1**
	b		*Present in both*: NH_2 ✓ CONH ✓ C=C ✓ COC ✓ *Present in zanamivir only*: COOH **and** OH ✓	*For similarities award **[2 max]** for any three correct, **[1 max]** for two correct, **[0]** for one correct.* *Accept C=N.*	**3 max**
	c		exposure of viruses to the drug favours resistant strains ✓ resistant strains difficult to treat ***OR*** drugs should be used only when required ‹not as prophylactic› ✓	*OWTTE*	**2**

(Question 14 continued)

Question			Answers	Notes	Total
	d		design chemicals to be less hazardous to health and environment ✓ use solvents/reagents that are less hazardous to the environment ✓ design ‹synthetic› processes that use less energy/materials ***OR*** design ‹synthetic› processes with high atom economy ✓ use renewable resources ***OR*** reuse/recycle materials ✓ treat waste to make less hazardous ✓ proper disposal of hazardous waste ✓		**3 max**

15.	**a**		$Mg(OH)_2(s) + 2HCl(aq) \rightarrow MgCl_2(aq) + 2H_2O(l)$ ✓		**1**
	b		n(HCl added) $= \langle 0.02500 \times 0.125 = \rangle 0.00313 \langle \text{mol} \rangle$ ✓ n(HCl unreacted with tablet) $= n(\text{NaOH}) = 0.00500 \times 0.200 = 0.00100 \langle \text{mol HCl excess} \rangle$ n(HCl reacted with antacid) $= \langle 0.00313 - 0.00100 = \rangle 0.00213 \langle \text{mol} \rangle$ ✓ neutralizing power $\langle \text{mol g}^{-1} \rangle = \langle \frac{0.00213}{0.200} = \rangle 0.011 \langle$ mol HCl neutralized per g antacid $\rangle$ ✓		**3**